

Practice Quiz: Communication

Name: __ Date: __

Part A: Multiple Choice

1. Communication in Active Inference is best described as: A) Sending information from a sender to a receiver B) Coupled inference — agents actively aligning their generative models C) Copying beliefs from one brain to another D) A one-way transfer of data
2. Theory of Mind is: A) A theory about what the mind is made of B) The ability to attribute mental states (beliefs, desires) to other agents C) The theory that the mind is separate from the brain D) A philosophical argument about consciousness
3. Language works as a communication tool because: A) Words have fixed, objective meanings B) Words activate similar patterns in the generative models of speaker and listener C) Language was designed by an intelligent creator D) All languages are identical
4. Cultural norms in Active Inference are: A) Arbitrary rules imposed by authority B) Shared prior expectations that reduce prediction errors in social interaction C) Universal laws of nature D) Only relevant in traditional societies
5. Echo chambers arise because: A) People are stupid B) Agents tend to seek information that confirms their existing model (low prediction error) C) Social media companies control all information D) People enjoy being wrong
6. A child who covers their eyes and says "you can't see me" is demonstrating: A) Mature Theory of Mind B) An immature Theory of Mind — confusing their own perceptual state with others' C) Good hide-and-seek skills D) Visual processing problems
7. Sarcasm is difficult to detect because: A) Sarcastic people are dishonest B) It requires inferring that the speaker's actual belief is opposite to their literal statement — a high-level Theory of Mind inference C) Sarcasm doesn't exist in some languages D) It requires hearing the tone perfectly

Part B: Short Answer

1. Explain how a misunderstanding between two people can be analyzed as a failure of coupled inference. Give a specific example and trace the prediction errors on both sides.
2. How would Active Inference explain why children under age 4 typically fail the "false belief" task (e.g., the Sally-Anne test)? What is their generative model missing?
3. Analyze misinformation using Active Inference: Why is it easier to believe false news that aligns with your existing views? What would need to change in a person's generative model to make them more resistant to misinformation?